FOAI - Assignment 2

Q.1 Chess Agents : Deep Blue v/s Garry Kasparov

PEAS Description

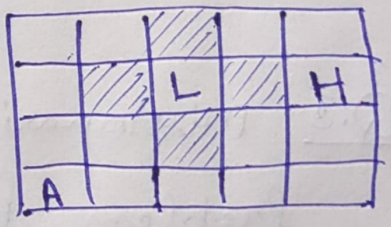
- Performance : Win, Draw, Lose, Time Management
- Environment : 8x8 chessboard, 32 pieces, chess clock, opponent.
- Action : Mechanism to move the chess pieces and press the clock (robotic arm)
- Sensors : Cameras (to sense chess pieces, and clock time).

Environment Properties

- Fully Observable
- Deterministic
- Sequential
- Semi-Dynamic
- Discrete
- Multi-agent.

Q.2 Grid World Agent (Agent, Lion, Home).

State Space Design



- State : Agents Coordinates (x, y) in $n \times m$ grid.
- Static Entities : Lion (x_L, y_L), Home (x_H, y_H)

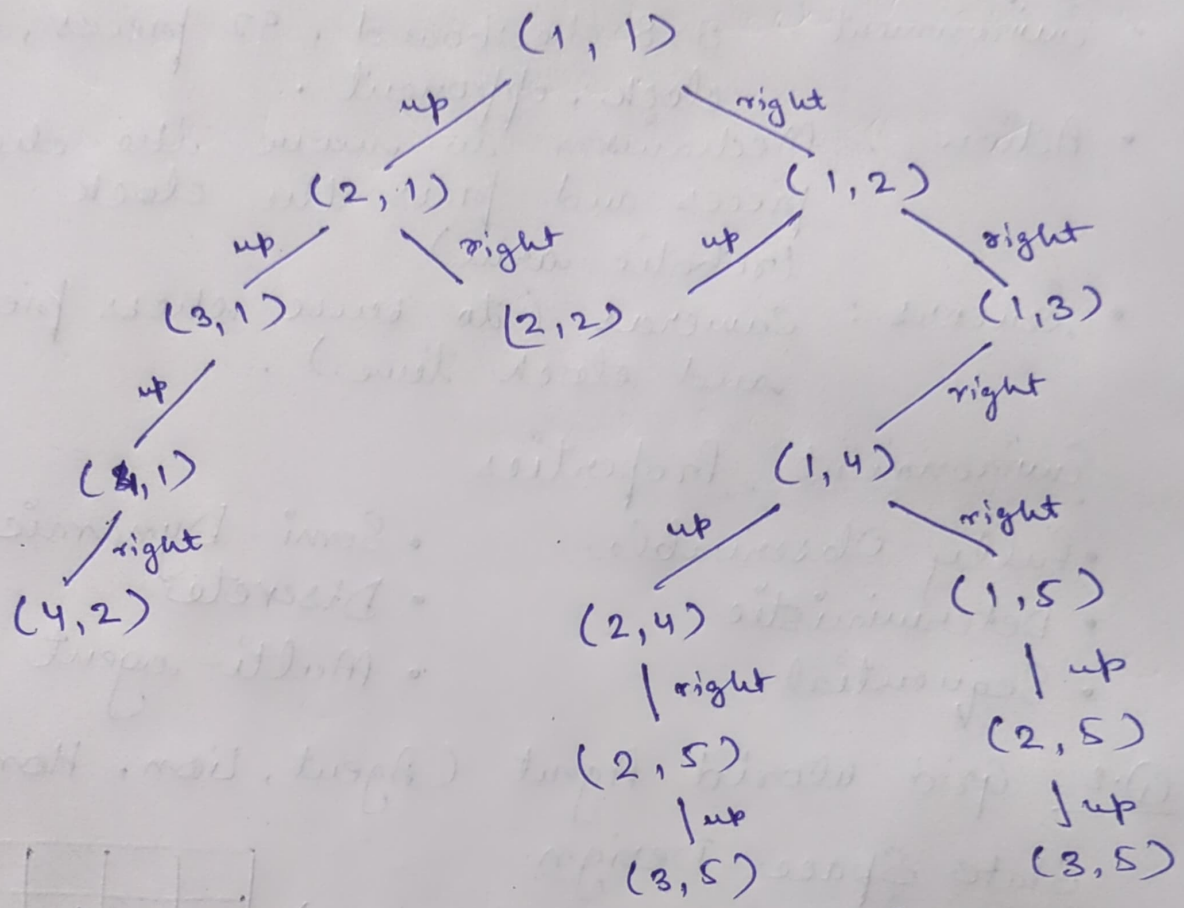
Precise Movement & Game Conditions

- Up : $(x, y+1)$ if $y < m$
- Right : $(x+1, y)$ if $x < n$
- Down : $(x, y-1)$ if $y > 1$
- Left : $(x-1, y)$ if $x > 1$

- Lose (Heat) - Agent loses if it enters a cell adjacent to the lion.

- Win (Home) - The agent reaches home.

Let's suppose,
Start point (Agent) is (1,1)
End point (Home) is (3,5)
Lion at (3,3)
so heat cells are (4,3) (2,3) (3,2) (3,4)



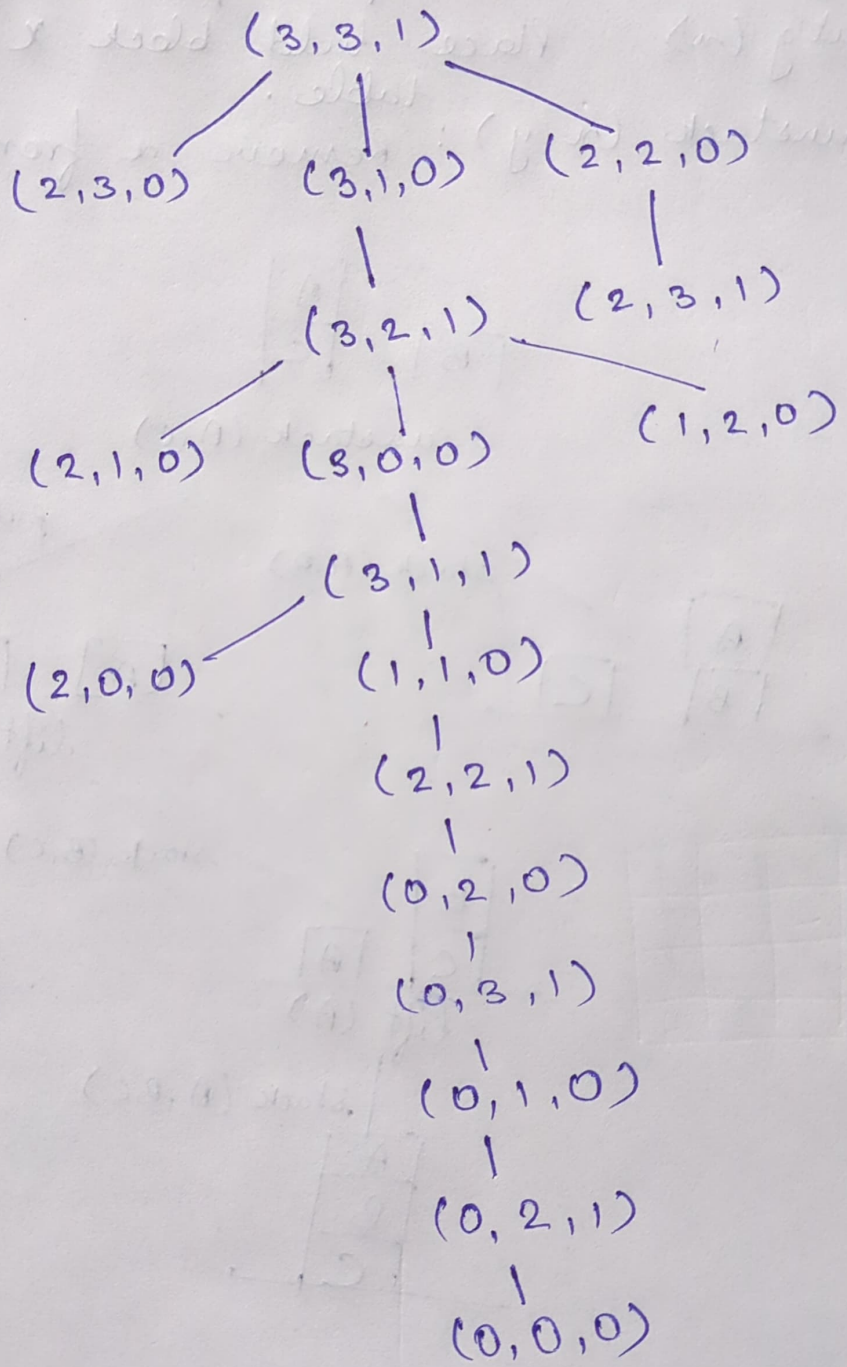
Q.3 Missionaries and Cannibals Problem

Problem formulation

- State Representation : (M, C, B) representing the number of Missionaries, Cannibals and Boats on the starting bank.
- Initial state : (3, 3, 1)
- Goal State : (0, 0, 0)
- Operators : Move 1 or 2 people across the river. Capacity k : $1 \le m + c \le 2$

- Validity Constraint : Missionaries cannot be outnumbered on either bank. for the starting bank : $M \geq C$ or $M = 0$

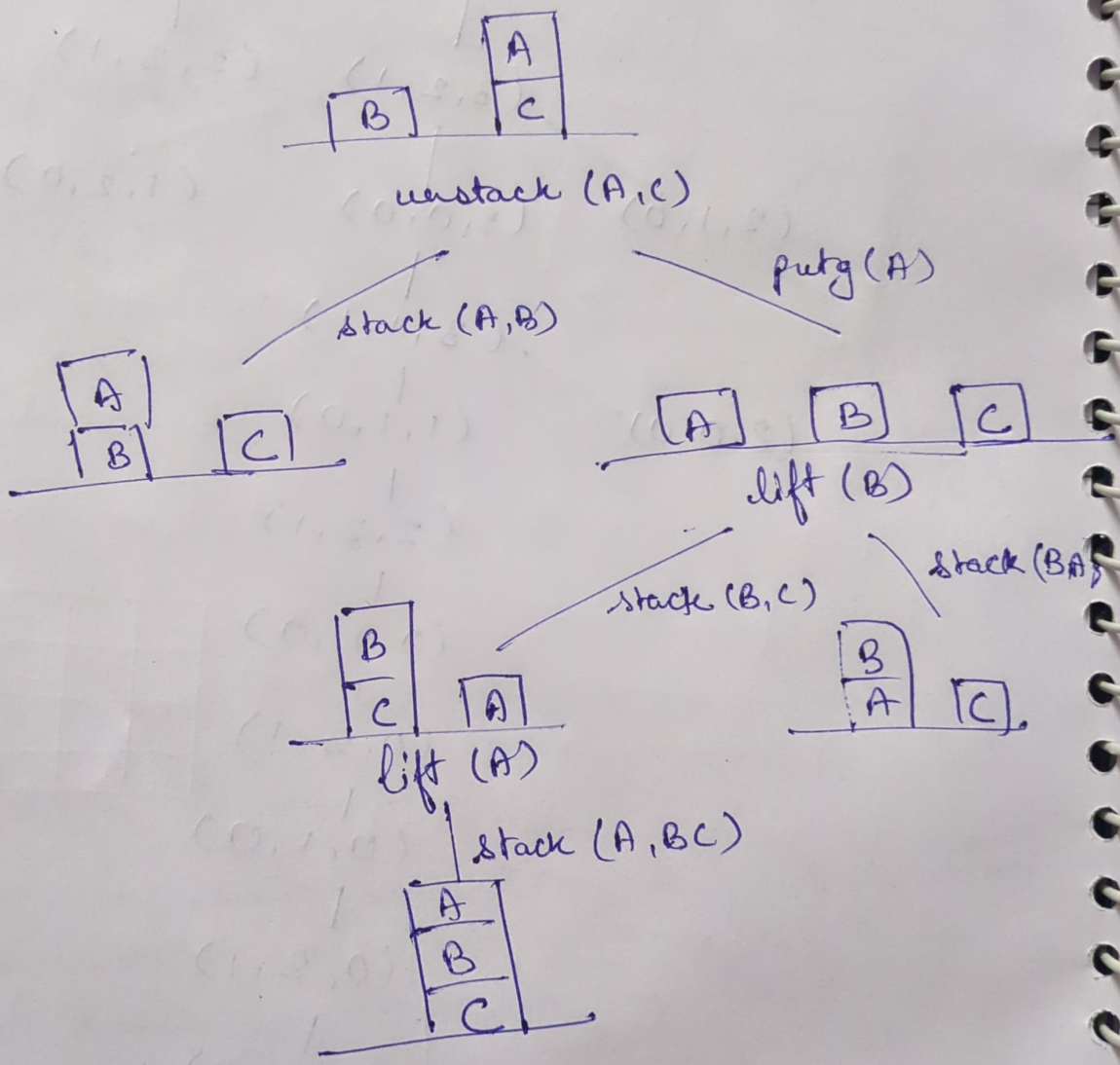
for the destination bank : $(3-M) \geq (3-C)$ or $(3-M) = 0$



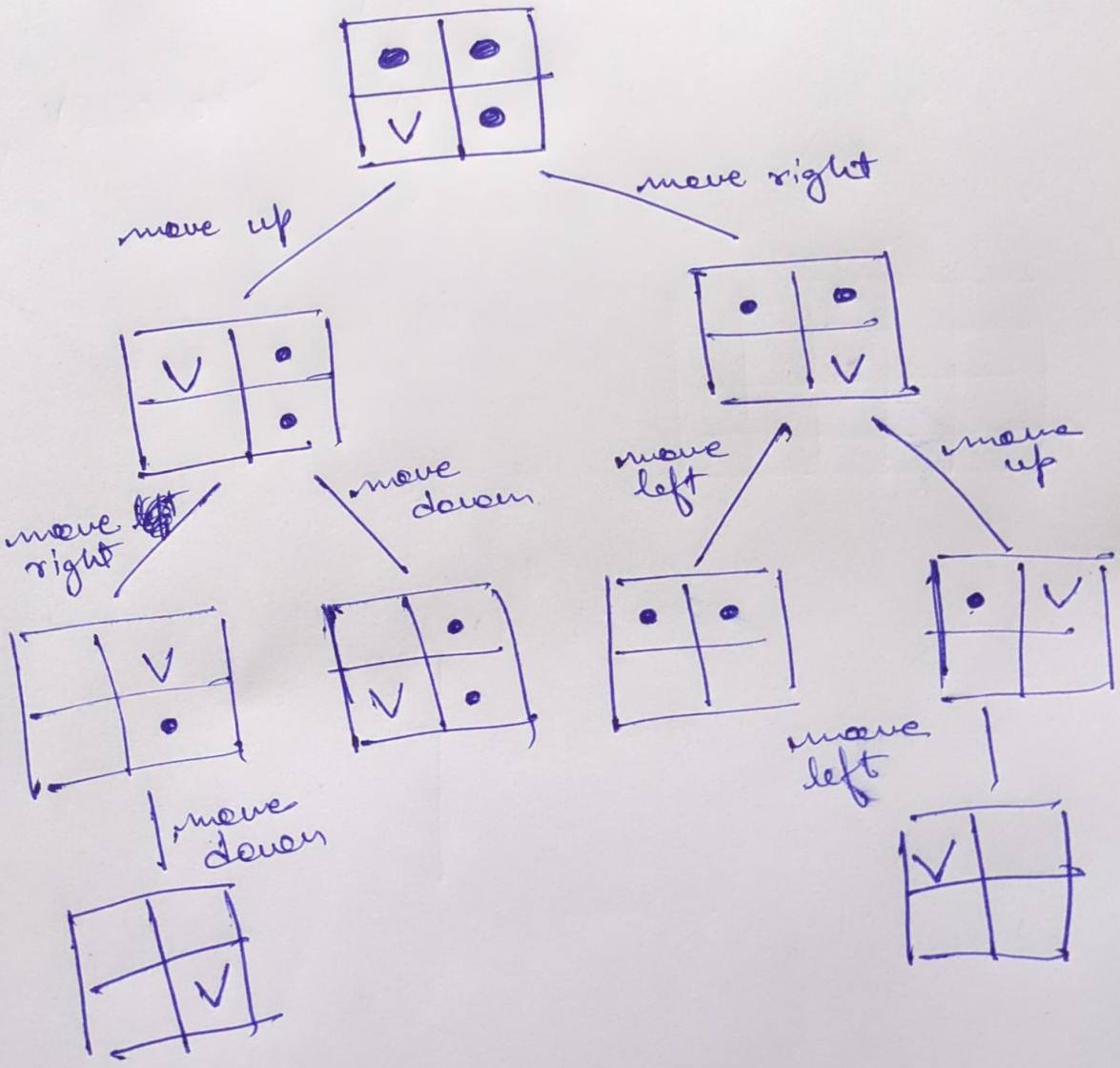
Q.4 Block World Problem

move rules -

- stack (x, y) : Place block x on y
- lift (x) : Pick up x from table
- putg (x) : Place held block x on the table.
- unstack (x, y) : Remove x from y



Q.5 Vacuum Cleaner Agent (2x2 Grid)

- PEAS Description —
  - Performance : Clean square
  - Environment : 2x2 grid, 4 locations
  - Actions : Move left, Right, Up and Down, Clean.
  - Sensors : Location sensor and Dirt sensor.

- State Space Properties —
  - Agent Positions : 4 possible cells.
  - Total states : $4 \times 2^4 = 64$ states.